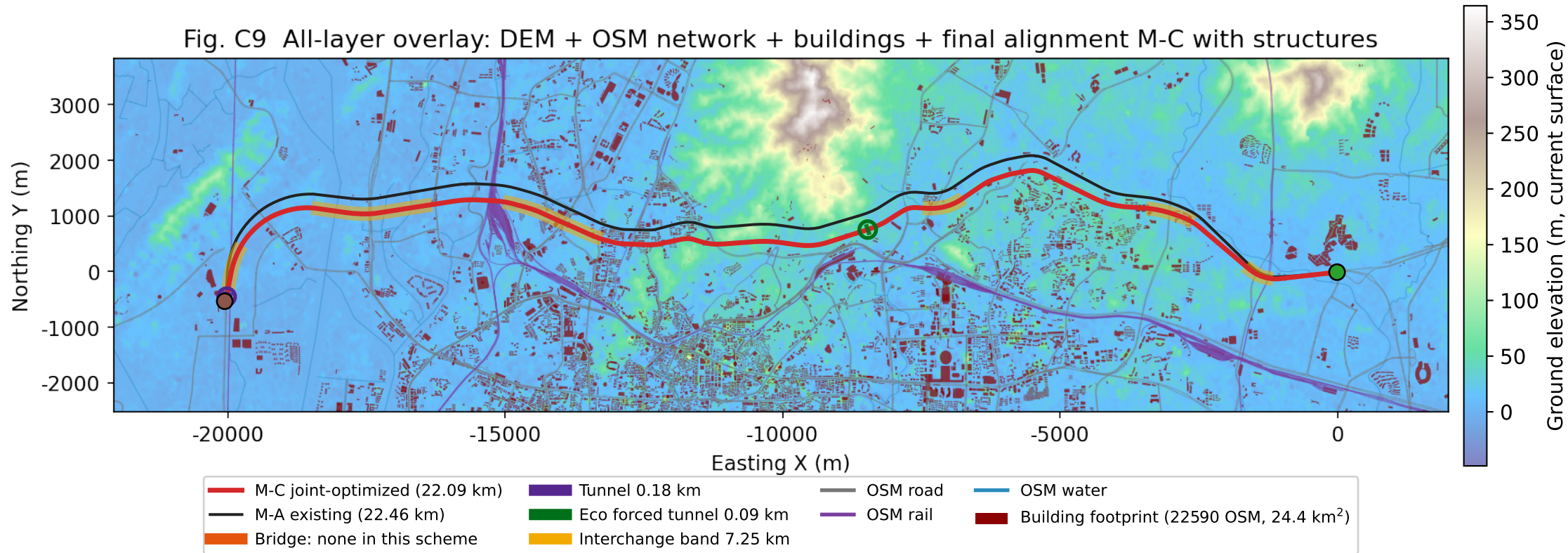


Fig. C9 All-layer overlay: DEM + OSM network + buildings + final alignment M-C with structures



Bridge/tunnel criterion identical to cost model (objective.earthwork_cost): earthwork cost > structure unit price, or fill>30 m / cut>30 m; exempt bands (eco tunnel, interchanges) excluded and drawn separately.